

Measuring AI Patent Enforceability with Large Language Models: Evidence from *Recentive*

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Abstract

On April 18, 2025, the U.S. Court of Appeals for the Federal Circuit held in *Recentive Analytics, Inc. v. Fox Corp.* that patent claims merely applying established machine-learning methods to a new data environment—without disclosing improvements to the models themselves—are ineligible under 35 U.S.C. Section 101. The Supreme Court denied certiorari on December 8, 2025. This proposal treats *Recentive* as a within-AI legal boundary that exogenously reduces the expected enforceability of applied-ML claims. The central question is whether this reduction leads banks to reprice and redesign corporate debt contracts for AI-patenting firms.

The empirical design builds a patent-level *Recentive* vulnerability score for 100 thousand AI patent claims. Instead of forecasting legal validity, the score uses a large language model to implement the Federal Circuit’s doctrinal distinction by analyzing textual similarity to vulnerable claim language. The patent-level scores are aggregated to a pre-period firm-level exposure measure for U.S. public firms. This proposal then links this exposure to DealScan syndicated loan contracts to investigate the joint modification of price and non-price specifications such as spreads, covenant structure, collateral requirements, maturity, facility size, IP-specific reporting and negative-pledge clauses. Identification leverages a DiD around the decision, which is complemented by a within-borrower-within-lender setup and predetermined-heterogeneity checks based on patent-pledging history, asset tangibility, and refinancing needs.

The pilot implementation shows that the classifier assigns high vulnerability scores to the four patents litigated in *Recentive* and generates economically interpretable firm-level exposure measures.

Keywords: AI Patents; Intangible Collateral; Trade Secrets; Bank Lending; Debt Contract Architecture

1 Introduction

A central insight of modern corporate finance is that the financing of intangible-intensive firms depends on the legal enforceability and pledgeability of intellectual property as collateral (Mann, 2018; Hochberg, Serrano, and Ziedonis, 2018; Horsch, Longoni, and Oesch, 2021). Existing work has shown that stronger patent collateral can expand debt capacity, while patent invalidation can reduce leverage. Much less is known about how banks adjust loan contract design when a class of intangible knowledge becomes harder to protect through patents and more dependent on secrecy-based protection. The recent ruling in *Recentive Analytics, Inc. v. Fox Corp.* provides a clean setting to study this question in the context of machine-learning technologies. In *Recentive*, the Federal Circuit held that patent claims merely applying established machine-learning methods to a new data environment, without disclosing improvements to the machine-learning models themselves, are ineligible under Section 101. The Supreme Court denied certiorari on December 8, 2025, leaving the Federal Circuit's precedential ruling in place. The decision changes expected enforceability within the AI patent universe.

Motivated by this shock, our main research question is whether the decrease in expected enforceability encourages banks to rethink corporate loan terms for AI-patenting firms. If applied-ML patents are less reliable as collateral, lenders may change more than just loan spreads. We accordingly look at a wide range of contractual responses including as collateral requirements, covenant packages, maturity, facility size, IP-specific reporting terms, and monitoring rights. This makes the setup useful for studying the value of intangible collateral as it contracts.

This study uses the large language model DeepSeek-Reasoner and structured legal reasoning prompts to construct the Recentive vulnerability score of artificial intelligence patents and summarise them into previous firm-level risk exposure indicators. The score does not predict the absolute legal effect of the patent, but measures its similarity with the text of the impacted claim in the *Recentive* case. We match this exposure with the DealScan syndicated loan data and set a double difference model around the *Recentive* judgement. It is expected that firms with higher prior exposure will face higher loan spreads, stricter restrictive terms and supervision requirements, and present the alternative effect of transferring to trade secrets.

2 Literature Review and Hypothesis Development

2.1 Patent Collateral and Intangible-Capital Leverage

Patent collateral research shows that intellectual property rights can be used as a support for borrowing when the lender can identify, evaluate and enforce the rights of the underlying assets. A study by Mann (2018) shows that 38% of U.S. patent application companies have used patents as collateral, and the court's judgement to strengthen creditor rights has increased debt issuance and R&D investment. Hochberg, Serrano and Ziedonis (2018) records an active secondary market for patent collateral, which supports risk loans. Horsch, Longoni and Oesch (2021) used the random distribution of judges of the Federal Circuit Court of Appeals (CAFC) to show that patent invalidation will significantly reduce the company's leverage.

These studies motivate our focus on the collateral channel, but leave open how lenders redesign loan contracts after a doctrinal shock changes the expected enforceability of a specific class of AI patents. *Recentive* affects the anticipated legal value of applied-machine-learning claims before most are litigated.

This setting enables us to study how a reduction in expected enforceable collateral value enters comprehensive contract design, including pricing, covenants, and monitoring

2.2 AI Patents and Lost Intellectual-Property Protection

Apart from the collateral channel, patents have well-documented effects on subsequent innovation (Sampat & Williams, 2019), suggesting that changes in patent enforceability may have economic consequences extending beyond the firm's immediate financing options. Acikalin, Caskurlu, Hoberg, and Phillips (2022) use large language models to construct firm-level *Alice* exposure, finding that affected firms increase research and development, secrecy, and non-disclosure-agreement usage. Helmers and Love (2025) study patent protection and software firm financing, but the debt-contracting implications of AI-specific enforceability shocks remain largely unexplored.

2.3 Artificial Intelligence and Corporate Outcomes

Recent research in AI and finance examines firms' exposure to artificial intelligence through labor, product markets, and technology utilization. Babina, Fedyk, He, and Hodson (2024) estimate AI investment using employee resumes and find that AI-investing companies grow through product innovation. Eisfeldt and Schubert (2025) use the release of ChatGPT to create generative-AI exposure and find a significant equity revaluation. Kogan, Papanikolaou, Schmidt, and Seegmiller (2023) link patent text to occupational tasks to construct labor-saving technology exposure.

2.4 Hypotheses

Building on the literature reviewed above, this study develops three testable hypotheses.

H1 (Price Channel). Following *Recentive*, firms with higher pre-period *Recentive* Vulnerability Scores face higher all-in drawn spreads on newly originated syndicated loans. The spread increase follows from the decline in the enforceable collateral value of affected AI patents. This hypothesis builds on Mann (2018) and Horsch et al. (2021) by predicting that debt pricing adjusts once the legal validity of the underlying collateral is impaired.

H2 (Collateral-Package and Control-Rights Reconfiguration). Higher exposure to *Recentive*-vulnerable claims leads to tighter non-price loan terms, including more stringent collateral requirements, stronger IP-specific reporting, negative pledges, IP-transfer restrictions, shorter maturities, smaller facilities, and tighter covenants. Building on Chava and Roberts (2008), lenders demand these enhanced control provisions as they update their assessment of borrower asset-substitution risk and the ex-post verifiability of intangible collateral. We predict no directional change in the share of IP-collateralized loans, as banks may either require tighter IP arrangements with stronger monitoring or shift toward tangible alternatives. Both responses represent reconfigurations consistent with our proposed mechanism.

H3 (Asset-Form Substitution). Firms with greater exposure to vulnerable applied-ML patent claims are expected to exhibit partial substitution from patent-form protection toward secrecy-form protection. Observable responses include more secrecy-related disclosures in 10-K filings, narrower claim drafting, and shifts in AI inventor retention, NDAs, and security hiring. This prediction combines the trade-secret protection channel in Klasa et al. (2018), the secrecy response to lost IP protection in

Acikalin et al. (2022), and the Myers and Majluf (1984) intuition that firms substitute across information and protection instruments when the relative cost of one instrument changes.

3 Data and Measurement

3.1 Grant Patent Data

This proposal uses the public PatentsView database to construct the universe of U.S. utility patents granted between 1990 and 2025. The patent-level data retain patent_id, application_id, grant_date, assignee_id, cpc_subgroup_id, examiner_id, inventor_id, and the full text of independent claims. The baseline AI patent universe is constructed using the U.S. Patent and Trademark Office's Artificial Intelligence Patent Dataset, 2023 update, by Pairolero et al. (2025). AIPD 2023 identifies AI content among U.S. patent documents published from 1976 through 2023, including both granted patents and pre-grant publications. For patents granted between 2024 and 2025 that fall outside AIPD's coverage window, the proposal supplements AIPD with PatentsView CPC class G06N, computational arrangements based on specific computational models, and keyword-based screening, followed by manual validation. Patent ownership is constructed as of the pre-Recentive date using the most recent patent assignment records observable on or before April 17, 2025, rather than relying solely on the original grant assignee.

3.2 The Recentive Vulnerability Score

The core measurement component is an LLM-based classifier that operationalises the Federal Circuit's *Recentive* test as a continuous Recentive Vulnerability Score on [0,1]. The score is defined as an indicator of the similarity between the text and the recent vulnerable claim language, rather than a prediction of how the court will rule on a specific patent. The current pilot uses DeepSeek-Chat and DeepSeek-R1 series, and the prompt is as follows:

“You are a patent attorney applying the Federal Circuit's two-step *Alice* test as articulated in *Recentive Analytics, Inc. v. Fox Corp.* Read the patent material below and output a score from 0.00 to 1.00 reflecting the degree to which this claim resembles Recentive-vulnerable applied-only ML claim language. Anchors: 0.00 = the claim recites a clear technological improvement to the machine learning model itself (a new architecture, loss function, training procedure, or representation); 0.50 = mixed or ambiguous; 1.00 = the claim recites only the application of established machine learning techniques to a new data environment, business problem, or domain, without disclosing a technological improvement to the model. Provide the score, a confidence rating, and a one-sentence justification quoting the specific claim language that drove the decision.”

The baseline classifier scores the first independent claim, with claims-plus-abstract and claims-plus-abstract-plus-specification configurations as robustness checks. The firm-level treatment intensity is the simple mean of the patent-level scores across the firm's pre-period AI patent portfolio:

$$\text{Vulnerability}_{i,2025Q1} = \frac{\sum_{p \in P_i^{AI}} \text{Score}_p}{|P_i^{AI}|}$$

where P_i^{AI} denotes firm i 's stock of AI patents granted before 2025Q1. The robustness plus total index covers the quoted weighted mean, the first quartile index, and the inventory and asset scaling

exposure. For details, please refer to Section 5.4. In order to eliminate the forward-looking bias, the ownership weight only uses the information available before April 18, 2025.

3.3 Loan Contract Data

DealScan syndicated loan data are used for the downstream business application. The main outcomes include all-in drawn spreads, covenant terms, secured status, maturity, facility size, and IP-related monitoring provisions. These variables allow the firm-level *Recentive* exposure measure to be linked to bank debt contracting outcomes.

3.4 Firm Financials and Disclosure

Firm-level control variables are based on Compustat's annual and quarterly fundamental data, covering the company's size, leverage ratio, profitability, R&D intensity, market value book ratio and asset tatability. The study of stock market events revolves around April 21, 2025, that is, the first trading day after the judgement, with the CRSP daily rate of return as the official data source. For the SEC's 10-K file, this study obtains plain text through the 10-X corpus cleaned by SRAF, uses the large language model classifier to extract sentence-level confidentiality marks, and uses the dictionary method of Hoberg and Phillips (2016) for robustness testing.

3.5 Sample Construction

The primary sample is a panel of AI patenting U.S. public firms. A firm enters the sample if it is listed in Compustat and holds at least three AI patents granted before 2025 Q1. Patents are linked to public firms through the KPSS patent permno linkage and then mapped to gvkey using the WRDS CCM linking table. Restricting the main sample to AI patenting firms focuses identification on variation within the AI patent universe, rather than on comparisons between AI firms and non AI firms. Non AI firms are retained only as external placebo observations. The loan level sample consists of primary sample firms with at least one DealScan facility between 2018 Q1 and 2026 Q1.

4 Empirical Methodology

4.1 Baseline DiD

The baseline specification is conducted at the loan facility level. The main specification relates loan contract outcomes to the interaction between a borrower's pre-period *Recentive* exposure and the post-decision period.

$$y_{f,i,t} = \beta(\text{Vulnerability}_{i,\text{pre}} \times \text{Post}_t) + \Gamma X_{i,t-1} + \alpha_i + \delta_{j(f),t} + \mu_{k(i),t} + \varepsilon_{f,i,t}$$

where f indexes facility, i indexes borrower, t indexes loan-origination quarter, $j(f)$ denotes lead arranger, and $k(i)$ denotes Fama-French industry. Post_t equals one if the loan origination date is on or after April 18, 2025. The specification includes borrower fixed effects α_i , lead-arranger by quarter fixed effects $\delta_{j(f),t}$, and industry by quarter fixed effects $\mu_{k(i),t}$.

4.2 Triple-Difference Using Predetermined Heterogeneity

The collateral mechanism implies that the treatment effect should be stronger when patent collateral is more relevant to the borrower or the lender. To test this implication, the proposal estimates a triple-difference specification using predetermined firm and lender characteristics.

$$y_{f,i,t} = \beta_1(\text{Vulnerability}_{i,\text{pre}} \times \text{Post}_t \times Z_i) + \text{Lower Order Terms} + \alpha_i + \delta_{j(f),t} + \mu_{k(i),t} + \varepsilon_{f,i,t}$$

Z_i denotes a predetermined heterogeneity variable measured before the *Recentive* decision. The main variables are pre-period patent-pledger status, low tangibility, near-term refinancing need, and borrowing from an IP-specialised lender. Each indicator is fixed at the pre-period level and is therefore not contaminated by post-*Recentive* contract responses.

4.3 Within-Borrower-Within-Lender Refinancing Design

Lending relationships are an important determinant of loan contract terms (Bharath, Dahiya, Saunders, & Srinivasan, 2011). To exploit this dimension and absorb relationship-specific heterogeneity, a complementary design compares loans issued by the same borrower–lead-arranger pair before and after the *Recentive* decision. Because strict same-facility refinancings yield insufficiently many observations, the design employs all loans from the same borrower – lead-arranger pair, restricted to similar loan purposes where possible:

$$y_{p,f} = \beta(\text{Vulnerability}_{i(p),\text{pre}} \times \text{Post}_f) + \alpha_p + \eta_{c(f)} + \varepsilon_{p,f}$$

p denotes a borrower lead-arranger pair. α_p absorbs all time-invariant characteristics of the borrower lender relationship. This within-borrower-within-lender approach extends the identification strategy pioneered by Khwaja and Mian (2008), which exploits variation across loan transactions of the same borrower with different lenders to isolate supply-side effects, by additionally exploiting time variation around the *Recentive* decision within the same pair. The estimating sample consists of pairs with at least one loan in the pre-period from 2023 Q1 to 2025 Q1 and at least one loan in the post-period from 2025 Q3 to 2026 Q1.

5 Pilot Results by LLM

5.1 Pilot Implementation

A stratified random sample of 5,000 patents is drawn from the firm-level scoring universe (U.S. ML patents granted 2010–2023 to Compustat-listed firms). DeepSeek-Reasoner returns 4,970 valid scores covering 381 firms (125 with ≥ 5 patents). Temperature is set to zero for replicability.

5.2 Validation Against Litigated Recentive Patents

The classifier is benchmarked against the four patents litigated and ruled patent-ineligible in *Recentive Analytics, Inc. v. Fox Corp.* itself. Table 1 reports the scores. All four patents receive scores of 0.85 or higher with high model confidence (mean 0.90), and model justifications quote the specific claim language identified by the Federal Circuit panel, including phrases such as "iterative training and dynamic updating" and "without disclosing improvements to the machine-learning model architecture, loss function, or training procedure." Three of the four litigated patents are not classified as machine learning by AIPD's binary ML flag (predict50_ml = 0), with underlying continuous AIPD ML scores below 0.50.

Therefore, the Recentive vulnerability score portrays a completely different concept from the AIPD classifier: AIPD aims to identify patents whose technology core is machine learning, while the vulnerability score specifically strips out the claim text similar to the purely applied machine learning patents that have been invalidated under Section 101.

Table 1. Recentive Vulnerability Score Applied to Litigated *Recentive* Patents

Patent No.	Grant Year	AIPD ML Flag	AIPD ML Score	Vulnerability Score	Confidence
10,911,811	2021	1	0.878	0.85	High
10,958,957	2021	0	0.123	0.85	High
11,386,367	2022	0	0.272	0.95	High
11,537,960	2022	0	0.418	0.95	High
Mean				0.9	

Notes: This table reports by DeepSeek-Reasoner for Recentive Analytics, Inc. v. Fox Corp. The Recentive vulnerability score generated by the four patents in the case that were invalidated under Section 101. The AIPD ML mark adopts the binary indicators of Pairolero et al. (2025) to define whether the patented core technology is machine learning; the AIPD ML score is the continuous variable at the bottom. The vulnerability score system is calculated by applying the prompts in Section 4.2 to the first independent claim of the patent. The four classification judgements have high model confidence, and the reasons given by the model accurately refer to the text of the claim encircled by the Federal Circuit Court of Appeals.

5.3 Inter-Model Robustness

Table 2. Inter-Model Robustness: DeepSeek-Reasoner vs DeepSeek-Chat

Statistic	DeepSeek-Reasoner	DeepSeek-Chat
Mean	0.747	0.773
Standard deviation	0.347	0.231
25th percentile	0.85	0.85
50th percentile	0.95	0.85
75th percentile	0.95	0.85
Maximum	1	0.95
Valid scores	4,970	4,970

As shown in Table 2, this study adopts the preliminary test of DeepSeek-Chat reproduction to evaluate the robustness of the cross-model. The average value of the two is roughly equivalent, which is 0.747 and 0.773 respectively. However, DeepSeek-Reasoner shows a significantly higher degree of dispersion, with a standard deviation of 0.347 and a maximum value of 1.00, covering the entire 0 to 1 interval; in contrast, the standard deviation of the Chat model is only 0.231 and the maximum value is

0.95. The wider distribution of the reasoning model shows that when the claim text clearly points to an extreme, it can decisively give extreme judgement instead of converging to the middle value, which is very valuable for downstream empirical identification.

5.4 Cross-Firm Distribution and Three Exposure Measures

Through three complementary indicators, this study adds up the patent-level score to the firm-level risk exposure. The combination constitutes the simple mean value of the patent vulnerability score of the firm sample. The stock exposure system extrapolates the number of vulnerable patents with a score of not less than 0.7 to the total artificial intelligence patent stock of the firm. Asset scaling exposure is obtained by the expected number of vulnerable patents divided by total assets, which shows the relative importance of the legal impact on firm financing. As shown in Table 3, there is a significant difference in the firm ranking generated by the three indicators, reflecting a completely different economic essence: the composition of the portfolio is related to the perceived quality of the IP collateralised asset package, the stock exposure determines the US dollar valuation of the pledged IP asset pool, and the asset scaling exposure affects the overall financing flexibility of the firm.

Table 3. Top-5 Most-Exposed Firms by Three Alternative Exposure Measures

Rank	By Composition (Mean Score)	By Estimated Vulnerable Patent Count	By Asset-Scaled Exposure
1	Wells Fargo Bank, N.A. (0.96)	International Business Machines (10,948)	Digimarc Corporation (0.95)
2	Pinterest, Inc. (0.96)	Google LLC (5,245)	Groupon, Inc. (0.36)
3	Applied Materials Israel (0.95)	Microsoft Corporation (4,768)	Verint Systems (0.10)
4	Electronic Arts Inc. (0.95)	Amazon Technologies (2,434)	Fair Isaac Corporation (0.08)
5	EMC Corporation (0.94)	Facebook, Inc. (1,577)	International Business Machines (0.08)

Notes: This table lists the top five firms according to each risk exposure indicator, and the sample is limited to companies with at least five patents in the preliminary test. The composition indicator of the combination is the mean score of vulnerability; the expected vulnerable patent stock index is the ratio of high-vulnerability patents with a score of not less than 0.7 to the total artificial intelligence patent stock of the firm; the asset scaling indicator is derived from the expected number of vulnerable patents divided by the total assets, and the asset data is taken from the number of Compustat According to the fiscal year 2024 and in millions of dollars.

As shown in Figure 1, the average risk exposure distribution across firms is 0.71, the standard deviation is 0.29, and there is a great distribution density in both poles. firms that apply machine learning to specific business operations are highly concentrated in the distribution high-end, mainly covering the fields of financial insurance, retail platform, firm software and operation optimisation. On the contrary, firms that develop core machine learning technologies are concentrated in the low-distributed end, mainly

including companies in the fields of semiconductor and electronic design automation, national Defence aerospace and artificial intelligence hardware. This distribution feature is highly consistent with the underlying legal economic logic, which strongly supports the previous firm-level vulnerability as a stable and theoretical processing variable.

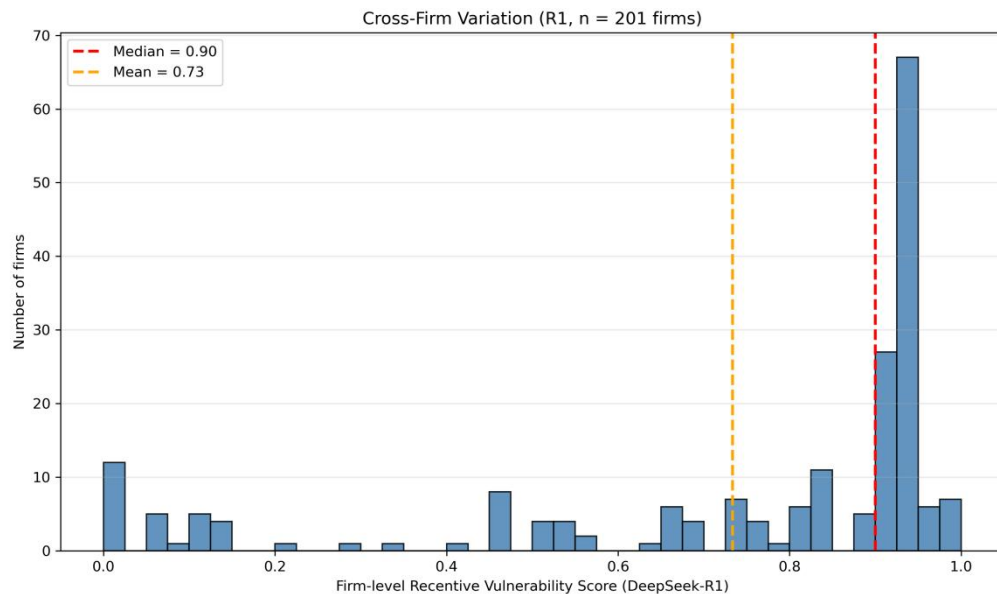


Figure 1. Cross-Firm Distribution of Firm-Level Recentive Vulnerability

Notes: Firm-level Recentive vulnerability score distribution of 381 U.S. listed companies with artificial intelligence patents. The score is calculated according to the average patent-level score of each firm in 5,000 hierarchical sampling samples. All vulnerability scores are based on the DeepSeek-Reasoner model.

5.5 Interpretation

The pilot results provide solid empirical support for this study. The measurement framework based on the large language model has successfully identified the typical patents impacted by the *Recentive* case. Its average score for the four litigation-related patents is as high as 0.90 and has a very high degree of confidence, establishing the feasibility of extending the framework to more than 76,000 full-sample patents. At the same time, the firm-level cross-sectional difference in the empirical distribution is fully in line with the theoretical expectations: applied machine learning firms show high vulnerability, while core technology firms show low vulnerability, and this invisible boundary cannot be captured by traditional industry classification or existing AI investment indicators. In addition, the firm rankings generated by the three alternative exposure indicators do not overlap each other, which further reveals the legal boundaries set by the *Recent* case in the field of artificial intelligence, which is being transformed into firm-level economic risk exposure through multiple independent channels. The above preliminary evidence system verifies the reliability of the measurement framework and lays the foundation for the debt contract analysis in Section 4, which is the core empirical contribution to the subsequent expansion of this working paper.

6 Discussion and Future Work

This study substantially advances the existing literature from three dimensions. Based on the patent mortgage literature (Mann, 2018; Hochberg et al., 2018; Horsch et al., 2021), this study shifts the focus from the total amount of loans to the multi-dimensional structure of loan contracts, revealing that when a specific intangible capital When the enforceability of the expected product is impaired, the lender will make dynamic adjustments to the spread, restrictive terms and supervision requirements. Following the vein of weakening the literature on intellectual property protection (Acikalin et al., 2022; Helmers & Love, 2025), the Recent case has set up the "application and The uniqueness of the "core" makes the empirical identification of this study completely free from the constraints of cross-industry comparison. Focussing on artificial intelligence and financial cross-documentation (Babina et al., 2024; Eisfeldt & Schubert, 2025), this design jumps out of the traditional framework of labour composition and product market exposure, and combines the legal status of AI knowledge assets with The pledgeability is established as an independent firm financing channel.

The empirical measurement of this study still has limitations. Vulnerability scores essentially only reflect the similarity between the target patent and the text of the high-vulnerability claim, not the prediction of absolute legal effect. Relying on 5,000 local samples to estimate the firm-level exposure of large diversified entities, substantial noise is inevitably introduced. The follow-up expansion of this working paper will be to professionally mark 300 to 500 sub-samples through full sample measurement and the introduction of practising patent lawyers, and build a two-stage classifier in combination with AIPD generalised AI marking and vulnerability score to systematically solve the above problems. The natural extension direction of the research is to construct more refined processing variables and weight the prior vulnerability exposure with the proportion of high-vulnerability patents as active collateral in firm-guaranteed credit agreements. The data construction of this indicator will deeply integrate the security record in the transfer of USPTO patents and the text of the intellectual property guarantee agreement disclosed by the SEC. The improvement of this identification strategy accurately separated the direct mortgage channel from the generalised expected impact channel, which constitutes the core priority of the next stage of this project.

In summary, the *Recentive* decision provides a clean legal shock to the expected enforceability of AI patent claims in the post-Alice era. The proposal combines this shock, an LLM-based text-based measure of claim vulnerability, and U.S. syndicated loan data to investigate whether banks reconfigure loan spreads, collateral packages, covenants, and monitoring rights when a class of AI knowledge becomes less pledgeable. The pilot implementation validates the measurement framework and establishes the foundation for the working-paper extension.

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